

Transport Research Arena (TRA) Conference

Quantitative methods for train delay propagation research

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Abstract

Due to the increase in railway traffic, trains increasingly need to adhere to their scheduled path to avoid extensive propagation of delays. This paper presents the findings of a systematic literature review covering articles related to the delay propagation problems of trains. By searching both the Scopus and Web of Science databases, the content of 156 relevant papers was screened and analyzed. The different methods, including optimization, simulation, analytical methods, and data-driven methods, are reviewed. Then, key issues in establishing different models using these methods for the train delay propagation problem are addressed. This review provides a foundation for the different methods adopted to solve different aspects of train delay propagation problems, thus giving support to future research work related to train delay propagation problems.

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1. Introduction

Many railway systems are operating at or near full capacity as a result of increasing demand. Furthermore, many train routes are designed to pass through multiple regions, resulting in a high level of train interconnection. Under these conditions, a train delay can be spread over long distances and cause timetables to be disrupted and destination arrival/departure times to be delayed, resulting in missed transfers, cancelled trains, and increased travel time, all of which contribute to a reduction in train service quality.

An extensive literature has emerged to understand delay propagation and to help prevent significant losses for both railway operators and passengers. From the timetabling and dispatching perspectives, we explore these works on the tactical and operational levels of railway traffic planning and management.

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At the tactical level, emphasis has been placed on ensuring the timetable's flexibility and robustness (Kecman and Goverde, 2015). This is due to the fact that timetable defects are one of the root causes that can have an impact on the entire railway system, as the delay of one train will affect other trains downstream due to the interaction between the trains in the network. This prompted researchers to focus on the areas of the appropriateness of time supplement allocation and conflict-freeness in the timetable (Bandara and Ekanayake, 2003; Zhou and Zhong, 2007; Bešinović et al., 2016; Zieger et al., 2018; Meng et al., 2019). At the operational level, a few key issues, including delay recovery, conflict detection and resolution, timetable optimization, and quality of service improvements, are always top priorities (Wen et al., 2019). When there is a large disruption, the railway capacity is reduced, and the scheduled timetable becomes infeasible even after utilising the time reserves. To address the potential conflict, various rescheduling strategies, such as changing the planned order of the train at a critical location, altering the succeeding train arrival, departure, and running times of the initially scheduled timetable, must be implemented. Thus, conflict detection and resolution in real-time traffic management are widely studied in the train delay propagation literature (Carey and Carville, 2000; Zhuang et al., 2016; Sparing and Goverde, 2017; Karoonsoontawong and Taptana, 2017).

In this paper, we review studies on the quantitative techniques used to study delay propagation. We perform a systematic literature review that aims to answer the following research questions:

1. How to classify delay propagation research based on quantitative methods?
2. What are the characteristics of each quantitative method identified for studying train delay propagation?

This review paper is outlined as follows: Section 2 describes the methodology used for the literature review. Section 3 presents different quantitative models that have been used in delay propagation. Section 4 briefly discusses the different quantitative models. Section 5 presents the concluding remarks and future directions.

2. Method

In March 2021, we conducted a literature search on the Web of Science and Scopus. The scope of the study was limited to academic journal and conference papers written in English, with no limitations on publication years. In the first step, delay propagation related keywords like *secondary delay*, *knock-on delay*, and *reactionary delay* were combined with terms like *conflict*, *headway*, *meeting*, *passing*, and *overtaking* to capture train interactions. *Time** for timetable or timetabling was used to narrow down the focus to a timetable perspective. The search results from both databases were combined, and duplicates were removed, yielding a total of 201 papers.

To remove irrelevant articles, a set of exclusion and inclusion criteria were used in the second step. This study did not include articles that did not have full-text access as well as qualitative papers. Articles that focused on other disturbance factors rather than delay propagation from a timetable perspective were filtered out. An investigation of the full article was conducted when the abstracts were reviewed, and the relevance was unclear. There were 156 articles left at the end of this step.

3. Results

Various methodologies have been used to deal with delay propagation problems. Inspired by (Hansen, 2006, 2010; Milinković et al., 2013; Wen et al., 2019), we grouped the literature into four categories: analytical, optimization, simulation, and data-driven methods. The distribution shown in Figure 1 is based on the categorization of the 156 articles in the final sample. Double counting occurs when two different approaches are used in the same article.

3.1. Optimization

Optimization-based methods aim to find the 'best' solution to delay propagation problems. Most optimization models involve formulating problems into mathematical models, such as Mixed Integer Programming, Mixed Integer Linear Programming, and Mixed-Integer Nonlinear Programming. The problems can also be expressed in other forms, such as Job Shop Scheduling, Simulation, Max Plus Algebra, alternative graph, and time event graph. These models are then solved using either approximate methods like Tabu Search, Heuristics, Branch and Bound, Branch and Cut, and Meta-heuristics, or exact methods like Decomposition, Column Generation, SAPI, or commercial solvers like CPLEX and Gurobi.

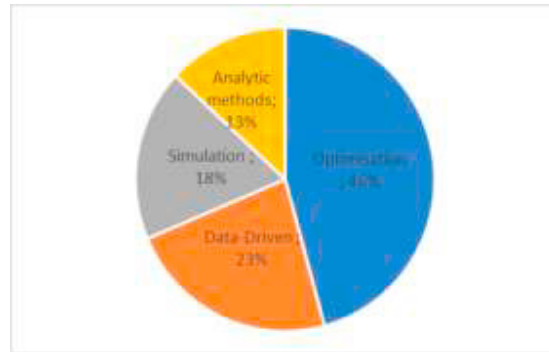


Fig. 1. Distribution of papers based on methods

Optimization-based methods are sometimes used to create optimal timetables that maximise capacity utilisation while minimising total network delays. Meng et al. (2019) propose a heuristic technique for fine-tuning an existing timetable in which the train schedule is modelled as a Directed Arc Graph, and a heuristic technique based on the critical path method is used to reallocate existing margins. Their findings reveal that adding time supplements to the appropriate positions in a timetable can reduce delay propagation and increase the timetable's robustness. Similar to buffer time allocation, optimum headway allocation (Bandara and Ekanayake, 2003) and running time allocation (Zhou and Zhong, 2007) have been implemented to optimise capacity allocation while avoiding proximal train interference and knock-on delay.

To optimise timetable rescheduling, train-related objective functions for optimization methods include minimizing deviations from the original timetable or reducing the number of trains cancelled. Some studies have used optimization techniques to resolve the conflict between trains (Carey and Carville, 2000; Sparing and Goverde, 2017; Karoonsoontawong and Taptana, 2017). The emphasis on rescheduling objectives has shifted toward a more passenger-oriented approach (Boland et al., 2012; Espinosa-Aranda and García-Ródenas, 2013), to minimise passenger delays. The most common strategies used to solve passenger-oriented train rescheduling problems include retiming (Li et al., 2019), reordering (Galapitige et al., 2018), rerouting, and even train cancellations (Buliček, 2018).

3.2. Simulation

Simulation approaches are used to model the railway system, with trains interacting with one another, with the infrastructure, and with passengers. Simulation-based methods are further classified as microscopic (RailSys, LUKS, OpenTrack, STRESI, or ATTPS), macroscopic (such as PROTON), and mesoscopic (such as PETER). The macroscopic model involves a higher level of aggregation, neglecting precise train dynamics and actual train separation (Keita et al., 2020). Macroscopic models, as opposed to microscopic models, require less data, take less time to compute, and are better suited to model large and busy networks. A mesoscopic model only provides details for certain parts of the railway network.

Aside from evaluating timetables, railway traffic simulation is used to assess the effects of various disruptions to scheduled traffic or the effectiveness of various dispatching and rescheduling options on bottlenecks, capacity utilization, and delay propagation. Tischer et al. (2020) assessed the capacity for specific train categories using a microscopic approach. Carey and Carville (2000) proposed microscopic models to assess the probability distributions of knock-on delays at stations when subjected to typical exogenous delays, such as changes in minimum headways, dwell times, or infrastructure changes. Bešinović et al. (2016) proposed a hierarchical framework for timetable design that integrates a network microscopic and macroscopic model. The macroscopic model optimizes a trade-off between minimum travel times and maximum robustness, whereas the microscopic model generates a feasible and stable timetable by iteratively adjusting train running and headway times. Luethi et al. (2009) used a microscopic simulation model to assess the effectiveness of a real-time rescheduling system in terms of reducing total knock-on delays and when new trains are added to the schedule. ROMA (Railway Traffic Optimization by Means of Alternative Graphs) is commonly studied in the literature for real-time rescheduling problems, where it is used to assist dispatchers in

overcoming their limited view of possible actions as well as ensuring optimal dispatching in perturbed dispatching traffic conditions (Corman et al., 2014, 2010).

3.3. Analytic methods (*Queueing theory*)

Analytical approaches refer to the queueing theory in this study. According to Li et al. (2020), analytic models tend to establish delay prediction models by computing the delay probability distribution in terms of Weibull, chi-squared, Erlang, exponential distributions, and log-normal based on the targeted data of train delays. This is consistent with Marković et al. (2015), who stated that analytic models compute the probability distribution of train delays as well as process durations such as running and dwell times based on historical data.

Analytical approaches, like simulation methods, can be used to evaluate operational performance. According to Milinković et al. (2013), queueing models are frequently used in the planning stage to assess the impact of increasing train frequencies, modifying infrastructure or train characteristics on the waiting time. Using queueing theory techniques, Weik et al. (2016) investigated the effects of traffic variability, such as train running stochasticity and dwell times, on the capacity of rail corridors. Without precise timetable data, the corridor is modelled as a series of highly correlated service stations, and the effective throughput, distributions of train running times, and service quality are calculated. Zieger et al. (2018) used the STRELE framework, which is based on a queueing- theoretic representation of railway lines, to investigate the effects of buffer time distributions on nominal capacity. Weik and Nießen (2020) also used the STRELE Formula to determine the permissible number of trains based on a quality-dependent capacity indicator and means knock-on delays calculated using an analytical queueing-based approach. Mu and Dessouky (2013) deduce the delay functions of the proposed switchable dispatching policy for a double-track segment using queueing theory techniques, and numerical experiments show that the delay functions produce accurate results when compared to simulation results.

3.4. Data-driven approach

The data-driven approach mines the relationship between variables embedded in the actual train operation data. This study examines three popular data-driven approaches, namely statistical models, graph- or network-based models, and machine learning models, as described in Wen et al. (2019).

3.4.1. Statistical methods

Statistical models such as correlation analysis, simple regression models, and visual methods are widely used to understand the distributional characteristics of train operations on specific lines or across the entire railway network. Statistical methods aggregate and summarise data in order to provide a broad picture or specific details on specific stations or trains, but they lack specific insight into recurrent delay patterns and the relationships between delays at different locations (Cerreto et al., 2018). Ulak et al. (2020) used kernel density estimation plots and cumulative empirical distribution function plots to compare the delay patterns of various routes and stations. Multiple linear regression was used by Gorman (2009) to predict total train running time while accounting for free-running time predictors and congestion-related factors. When measured in total train delay, the regression results reveal that each pass/overtake causes a significant net loss. Graffagnino (2013) introduced an alternative definition of a boxplot that accurately represents rail traffic data, as well as a top-down method for tracking the most serious traffic issues in terms of punctuality targets. These methods can be used to improve the timetable and track the effectiveness of the steps taken.

3.4.2. Graph methods

Graph-based models attempt to derive information from data in order to establish the arc weights of alternative graphs, the matrices of the Markov model and max-plus algebra, the Bayesian network's probability chains, and the fuzzy numbers of the fuzzy network (Wen et al., 2019). In graph-based models, train events like arrival, passage, and departure are represented as nodes on a graph, with their relationships expressed as joint probability density functions. The graph method relies on prior dispatching knowledge. To recover a schedule, Li et al. (2019) proposed the max-plus algebra method to recover a schedule in which six operational adjustment strategies are analysed to determine the least number of delayed trains. The feasible solutions are imported into the second model to optimise the operation

adjustment. Zhuang et al. (2016) presented two conflict prediction scenarios, one with no additional disturbances and the other with additional disturbances. The Bayesian network structure, which can express interactive and causal relationships between the different variables through graphical structures, is well suited to modelling the complexity, interdependence, and dynamical behaviour of train operations. The Bayesian network models were used by Huang et al. (2020) to predict the effects of disruptions and disturbances in terms of primary delay, the number of affected trains, and total delay times. Zhang et al. (2019) proposed a conditional Bayesian delay propagation model with Markov property to investigate delay propagation on the Sydney railway network.

3.4.3. Machine learning methods

Machine learning (ML) uncovers hidden knowledge by learning trends in historical datasets to produce a reliable and repeatable prediction (Wen et al., 2019). ML is further divided into supervised and unsupervised learning. Unsupervised learning uses unlabeled data to learn clustering patterns in the data, whereas supervised learning uses labelled data to make predictions. We find that the majority of the ML models developed were supervised models, either standalone or with prediction output used for subsequent decision support. Accurate delay prediction is critical at the tactical level for establishing a flexible and stable train schedule. At the operational level, accurate delay prediction can detect recurring delays on railway routes and potential conflicts in train operation. For train event based prediction problems (i.e., delays, running times, and dwell times), supervised learning models are commonly used. Marković et al. (2015) proposed support vector regression to capture passenger train arrival delays while taking infrastructure characteristics into account. To provide more accurate real-time predictions of travel times, dwell times, arrival delays, and departure delays on the UK's railway network, Taleongpong et al. (2020) developed the deep neural network and extreme gradient boosting models. On the other hand, unsupervised learning is used to uncover the patterns embedded in the train operation data. Huang et al. (2019) presented a data-mining approach that uses the K-Means clustering model and probability density fitting techniques to estimate the impact of high-speed train disturbances.

4. Discussion and Conclusion

In this paper, we discussed the classification of delay propagation literature based on the method used. The commonly utilized methods were analytical, optimization, simulation, and data-driven methods.

Optimization methods use deterministic variables to search for the optimal value of a particular objective function, such as minimal deviation from the planned schedule, under predefined constraints, such as minimal running times (Hansen, 2006). In other words, the optimization models strive to deliver optimal or near-optimal solutions for train delay problems by weighing the trade-off between two variables such as service quality, buffer time allocation, or capacity allocation. The drawback of optimization methods is that they tend to be dedicated to single scenarios and

seek global optimality rather than global viability taking into consideration all other constraints, as is customary in the railway industry.

Simulation methods do not have an objective function; instead, they mimic real-world railway system behaviour and offer little or no quantitative solution (Corman and Meng, 2014). This indicates that simulation methods do not deliver the best delay recovery solutions. Simulation is highly computationally intensive due to the necessity to execute multiple possible scenarios. Simulation models are complex since many different data types, such as infrastructure, passengers, and timetables, are required to describe the complex train system; often assumptions are made in the absence of specific types of data. A complex real-world issue could be oversimplified by the assumptions made, giving simulation results that do not fully reflect the actual railway system situation. Besides that, simulation is also impractical and time-consuming for larger railway networks.

Simulation and analytic methods are used to assess the effectiveness of scheduled timetables and dispatching strategies in addressing potential disruptions. For example, analytical and microsimulation approaches are often used to measure waiting times and capacity utilisation (Hansen, 2010). Analytical approaches, according to Milinković et al. (2013), involve simplified assumptions and do not require a large amount of input data. Although queueing theory is effective for estimating delays, it is unable to accurately distribute expected delays across the entire rail network because it overlooks the complex train network and interdependency between trains (Higgins and Kozan, 1998) and stochastic nature (Weik et al., 2016). Hansen (2006) also indicated that most queueing models assume that

the service periods and train intervals are independent random variables, which is debatable, particularly when dealing with periodic timetables with high frequencies.

Data-driven methods offer a practical and reasonable solution to train dispatching support since they are less computationally intensive by eliminating the need to run thousands of simulation model scenarios. Statistical analysis is used to uncover dependencies between related variables in train operational data. According to Milinković et al. (2013), statistical analysis can provide insights into train operation performance and can be used to supplement understanding of the rail system based on other methods such as simulation. This approach is appropriate for railway systems that are well-regulated and operate in stable conditions, but not for systems with many potential disruption sources. The graph theory is frequently used to solve conflict resolution problems. ML helps predict delay propagation more accurately as ML models are better at fitting nonlinear and high-dimensional data than statistical models. However, the results derived from ML models are less interpretable. The main drawback is that data-driven approaches greatly depend on the availability of good-quality and a sufficient amount of historical train operational data.

This study has some limitations. For example, we limited our search to journals and conference proceedings from only two digital databases. Furthermore, this study only includes articles that were published in English. Thus, there is a risk that articles in other languages relevant to this study were overlooked. Thus, it is proposed to extend this study in the future to include more digital databases as well as languages other than English for more wide-ranging and in-depth findings. This work identifies and gives an overview of the four methods employed in train delay propagation research. Thus, another extension of the study would be to investigate each identified method in greater depth and from a different perspective.

With the availability of big data, the data-driven method, especially ML, has attracted the attention of many researchers when studying delay propagation. This paper identified the existence of a research gap, that is, the potential expansion of ML approaches in other aspects such as buffer time allocation, capacity utilization, and optimizing the train performance besides predicting the train movements in terms of train arrival delays or running time. There also exists a great potential in developing the different types of robust hybrid model that adopts the promising prediction result from the ML model in problems related to delay propagation.

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